Shelter and Access: One-Sided, Falsifiable Interior Measures for Generated Voxel Buildings

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Abstract

Evaluation of open-ended building generation in embodied environments (whether the builder is a procedural generator, a diffusion model or an LLM agent) still rests on human ratings, on visual models, or on similarity to a reference solution. The first two yield no falsifiable quality claim about a single building, and similarity requires a reference solution that open-ended building does not have. None of them answers the question: can an agent with a body use this structure as a house? We propose the interior field Ω, defined by voxel geometry alone, which measures how enclosed each air cell is; it has no trainable parameters. On top of it we build two one-sided measures: *shelter* indicates whether a building can provide cover, and *access* indicates whether there exists an agent-centric path (one a player-sized body can walk) from relatively outdoor space to relatively indoor space. One-sided means the measure's house-quality claim is made in the low direction only and is falsifiable there: a low score means the building fails a necessary condition of a usable house (no cover, or no way in), a fact checkable in its geometry; a high score means only that the condition is met, not that the house is good. We also propose VoxHouse3D, a dataset of 2,007 player-built houses and 2,023 same-provenance non-house structures, taken from 28 archived worlds of real community servers, one map file each.

1 Introduction

Minecraft is the main testbed for embodied building-generation research [23]: the settlement generation competition has run for years [18]; after dialogue-enabled building agents [19], LLM agents began to build inside the game [17, 40]. Whether the goal is embodied AI or structure generation, the same gap remains: a falsifiable measurement of whether the produced structure supports embodied use.

None of the current evaluations reads the geometry itself and turns it into a checkable per-building claim. Human ratings are subjective and cannot be re-run; GDMC's four-axis human judging is of this kind [18]. LLM-as-judge [8, 9] carries position bias and self-preference, and moreover reads only an exterior rendering: it can score a build near the top even when the render already shows the defect (Figure 1), and an interior defect is not in the picture at all (Figure 4). Distributional similarity such as the FID family [10] treats resemblance to the training set as good and says nothing about a single building. Task-based embodied benchmarks have programmatic criteria, but they need an enumerable goal or a reference solution, and open-ended building has neither; isovist and visibility-graph analysis from architecture does not depend on learning, but its validity argument stops at corpus-level correlation and names no condition against which a single building could be checked [1, 2]. To our knowledge, no existing evaluation combines all three: requiring no reference

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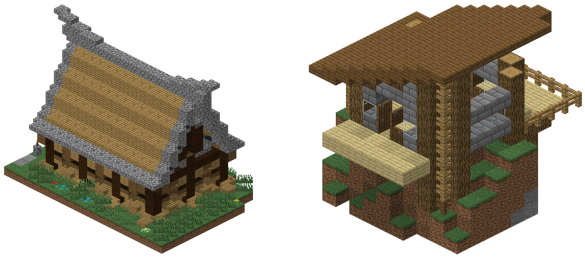


Figure 1: **What an exterior judge overlooks.** A multimodal model judging renderings of player-built houses gives all three scores (is a house / can be entered / provides cover, in that order) close to the top score of 10, yet each sample has a defect the render itself shows or suggests. This paper builds the interior field directly on the voxel geometry and reads out the two measures.

solution, reading the geometric structure directly rather than the exterior, and making a falsifiable quality claim about a single building.

What evaluation practice is missing is not more scores but a geometric measurement that is falsifiable about a single building and whose low direction has an identifiable meaning. This paper sits in that gap, on the discriminative side of 3D modeling of generated structures. We propose the interior field Ω: a discounted multi-bounce field defined by voxel geometry alone, which measures how thoroughly the building encloses the air inside it, with no material, no photometry, and no trainable parameter of any kind. On top of it we build two one-sided measures, shelter and access. One-sided means the quality claim is made, and is falsifiable, at low scores only: a low score is a failed necessary condition of a usable house; a high score means only that the geometric structure meets the structural condition, not that the house is good. Our contributions are:

- The interior field Ω and two one-sided measures: a discounted multi-bounce field with an exact sparse linear-system solution (no sampling, no learning, no parameter fitted to labels), and the shelter and access measures defined on it.
- An evaluation protocol that reports alarm rates against a frozen reference threshold (the human-house 10th percentile), shipped with the package as `structgauge bench`.
- The VoxHouse3D dataset, taken from real community multiplayer servers with same-provenance negatives: 28 worlds, 2,007 player-built houses plus 2,023 non-house structures from the same servers.

2 Related Work

Judging paradigms and the one-sided guarantee. Approaches to evaluating agent outputs come in three kinds: programmatic verification, LLM/VLM judging, and human rating. The benchmark for the programmatic kind is SWE-bench, whose patch criterion is falsifiable and one-sided: failing the tests means the code is wrong, while passing does not mean there are no bugs [36]; the programmatic criteria of task-based embodied benchmarks [35] and similarity comparisons [34] belong to the same kind, but these methods require a reference solution or an enumerable definition of success, and open-ended building generation has neither: it has no unique correct answer. Open-ended building therefore falls back on judges: GDMC's human rating, and Creative Agents using GPT-4V as the evaluation measure [32]. The one-sided form has an older lineage: soundness in static analysis gives a guarantee in one direction only [24], soundiness argues for stating the practical deviations explicitly [25], and red-teaming actively hunts for the places where a claim fails [26]. In contrast to the evaluation approaches above, we fill the gap with a programmatic criterion that needs no reference solution and no enumerable goal, guarantees quality only in the low direction, and remains refutable for a single building.

Minecraft and PCG evaluation. The GDMC settlement competition is judged on four human-rated axes [18]; MineAnyBuild [17], which scores with a VLM instead, sits at a different layer of the pipeline. Automation of geometric analysis has two strands: Hervé and Salge correlated PCG measures with human ratings [2], and the same authors also pointed out the problem of unstated judging criteria [16]; isovists have been automated in Minecraft [1]. Existing tooling also includes

expressive-range analysis [11] and a taxonomy and survey of evaluation practice [12]. Unlike these works, which establish validity at the corpus level, we push the validity argument to a falsifiable claim about a single building.

Geometric measurement. This lineage has four strands. The first uses a single line of sight: isovists and visibility-graph analysis [4, 5] quantify spatial openness, as does the sky-view factor of urban canyons [20]. The second simulates light: ambient occlusion and obscurance [7], the discrete-ordinates method of radiative transfer [6], and radiosity [27] model multiple bounces of light. The nearest lineage is indoor-scene evaluation, which has incorporated reachability and physical interactability [28, 29], but with threshold-based rules on synthetic furniture scenes. Embodied spatial-reasoning benchmarks and platforms [30, 37, 38, 39] ask a different question: a model's spatial ability. Unlike these four strands, Ω has no material and no photometry; it is a construct field (built to measure a concept, not to simulate light), with a discounted exact solution for multiple bounces (no sampling).

3 Method

We propose a two-layer construction that does no rendering and reads the voxel geometry directly to judge a structure: the interior field Ω (Section 3.1), and two one-sided measures built on that same field: shelter tests whether the interior is completely enclosed (Section 3.2), and access asks whether a player-sized body can walk a route from relatively outdoor space into relatively indoor space (Section 3.3). We call a structure a usable house when it provides cover and can be entered; the two measures test these two necessary conditions. The whole pipeline is deterministic. A one-sided measure makes its falsifiable house-quality claim in the low direction only: a low score is a failed necessary condition of a usable house; a high score means only that the condition is met, not that the house is good. A good house can only be judged by higher-level criteria (Section 6.1).

3.1 The interior field Ω

The most basic notion of a house is that it needs interior space, and we make "how enclosed" a computable quantity: **looking out from an air cell, how thoroughly is the line of sight blocked?** Counting blocked directions is not enough: for a cell deep inside, almost every direction is blocked by a wall, and a single-bounce count cannot distinguish it from a sealed room; so we ask one step further: how enclosed is the air cell just in front of the wall that blocked it? (Figure 2)

The input is a voxel occupancy array. Each air cell x casts rays along the 24 directions of a level-symmetric S_4 quadrature set (all permutations and sign changes of one direction triple; uniform weights $1/24$); a ray that hits a solid block stops at the last cell y before the impact, is discounted by γ, and is re-emitted from y over the same 24 directions (the recursion that the field value v_y below denotes), until the mass escapes the bounding box (the sample's tight extent). Let h_x be the fraction of directions from x that hit a solid, and K_{xy} the fraction that stop at y, so that $\sum_y K_{xy} = h_x$. The value of every cell then satisfies $v_x = h_x + \gamma \sum_y K_{xy}\, v_y$, or in matrix form:

$$(I - \gamma K)\, v = h, \qquad \Omega = (1 - \gamma)\, v \in [0, 1]$$

A completely sealed air cell has $\Omega = 1$, a completely open one $\Omega \to 0$: Ω is a continuous notion of "how enclosed is this cell", with no threshold on entrance size and no need to decide a room partition. Ray traversal uses a supercover DDA (digital differential analyzer), and the field is invariant under all 48 octahedral symmetries (the choice of the tie rule and the empirical checks are in Appendix A). We set $\gamma = 0.90$; Section 6.2 gives the sensitivity analysis. Computation takes median 0.083 s per build on a single core (largest sample 182.2 s).

From field to measure. Both measures share one form: for each scoring unit, take a weighted average with Ω as the weight. They differ in the scoring unit and in the quantity averaged: a binary blocking event for shelter, a continuous Ω-range for access. Volume cancels in numerator and denominator, so the value is a proportion, not a total: scaling the building up, or drawing a larger bounding box (exterior air cells have $\Omega \approx 0$ and contribute almost nothing), leaves the value unchanged. This is the first line of defense against size confounding.

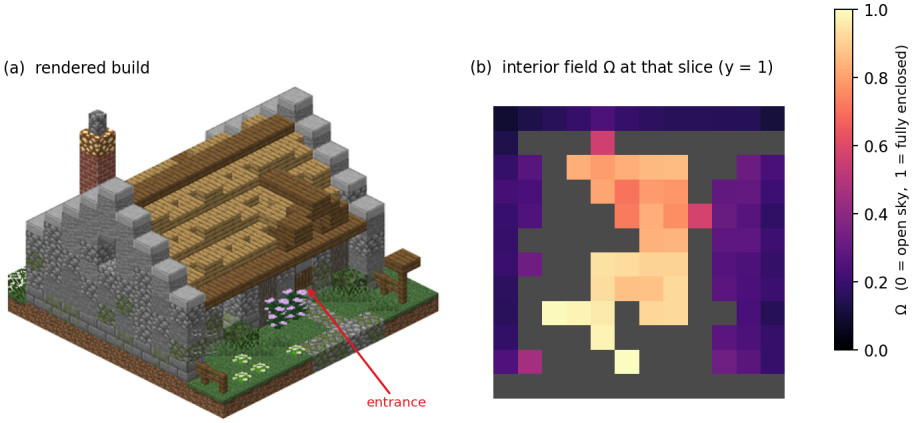


Figure 2: **A build and a horizontal slice of its interior field.** A player-built cottage with stone walls and a wooden roof (left; orthographic render) and the horizontal slice of its interior field on the walkable ground-floor layer $y = 1$ (right), sample `VH3D-w07-700de289` (13 layers, 870 scored cells; shelter 0.6194 / access 0.7726). On the slice the middle of the interior is close to saturation ($\Omega \approx 1$) and the field decays continuously towards the openings at both ends and the outer ring.

3.2 Shelter

Shelter asks: is the interior of this building really enclosed? The scoring unit is an air cell within the sample; the event is that a straight line cast from that cell upward and in each of the four horizontal directions (all five) hits a solid block before reaching the sample boundary ("no rain from above and no wind from any side"); the weight is $w = \Omega$. The value is the sheltered proportion of the Ω mass (the sum of Ω over the scored cells). The weight is Ω rather than uniform: a uniform weight would count the open cells around the house on a par with the interior cells, and the value would then track how much open ground surrounds the building; weighting by Ω lets the geometry itself decide what counts as interior.

A low score names one of two defects: no roof or open sides. In the low direction, shelter flags a structure that cannot keep out wind or rain.

3.3 Access

Access is the embodied measure: it asks whether a player-sized body can walk from relatively outdoor space into relatively indoor space. The scoring unit is a walkable component: the nodes of the walk graph are positions where a 1×2 player body can stand (support underfoot, and both the foot cell and the head cell passable); the edges are the 4 horizontal directions $\times\ \Delta y \in \{-1, 0, +1\}$, 12 neighbors in all; a climbing edge requires a climbable block at its endpoint, and a door is passable but cannot be stood on. The walk graph (Figure 3) is exactly what separates access from ordinary geometric connectivity.

The criterion is that a component touches both relatively outdoor (low-Ω) and relatively indoor (high-Ω) cells, i.e. a path from relatively outdoor space to relatively indoor space exists; access measures the weighted average of how large this indoor–outdoor gap is for each component (the Ω-range r_i); a component lying entirely in relatively outdoor space or entirely in relatively indoor space has r_i near 0. This value is not an added rule but follows directly from the definition of the interior field, so the measure needs no threshold or seed point to separate indoor from outdoor.

$$\text{access} = \frac{\sum_i m_i\, r_i}{\sum_i m_i}, \qquad m_i = \sum_{x \in \text{comp}_i} \Omega_x, \qquad r_i = \max_{x \in \text{comp}_i} \Omega_x - \min_{x \in \text{comp}_i} \Omega_x$$

The defect flagged in the low direction is that an agent cannot get in: completely sealed, fragmented walkable components, or little interior overall. Three extremes anchor what the measure means: a completely sealed box (Ω is identically 1 within the component, hence $r_i = 0$) contributes 0; a wall-less pavilion (Ω near 0 throughout the component) also contributes little; a house with a door and an interior (a component spanning from $\Omega \approx 0$ to $\Omega \approx 1$) reads high.

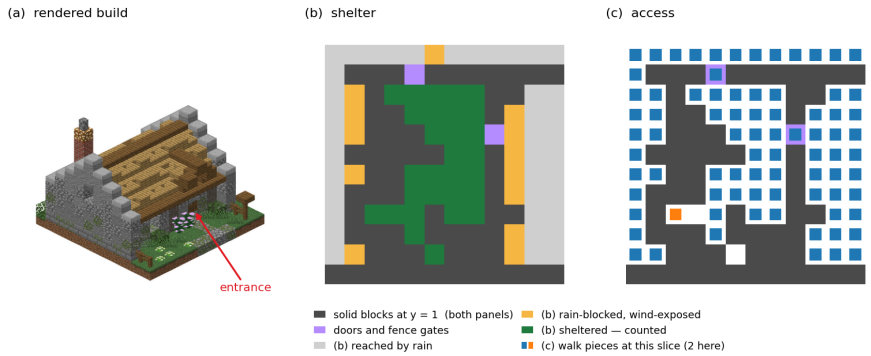


Figure 3: **What each measure sees on the same build** (`VH3D-w07-700de289`). (a) Orthographic render. (b) The per-cell exposure classification for shelter on the layer $y = 1$: reached by rain / rain-blocked but wind-exposed / both blocked (counted as sheltered): openings on both sides and at the top edge let wind into the interior, so only the middle counts towards shelter. (c) The walkable components for access on the same layer, one color per component (blue = the main component reachable from outside); components are computed in 3D, not per layer; doors and fence gates are marked in lavender. The main component covers the whole layer except one isolated cell in the lower-left corner. Whole-build values: shelter 0.6194, access 0.7726.

4 Dataset

VoxHouse3D contains 4,030 player-built structures segmented automatically from 28 archived map files (each a single server world): 2,007 houses and 2,023 non-house man-made structures. The closest existing data, 3D-Craft [31] (crowd-sourced, thirty-minute laboratory builds), has no negative class, so it cannot support house-vs-non-house discrimination. Our negatives are same-provenance: not synthetic structures from another distribution, but player-built non-house objects from the same maps (pixel art, rail systems, farms, statues); with shared tools and the same in-world environment, the remaining difference is mainly the structure itself.

A sample is a single connected player-built structure. The pipeline excludes naturally generated structures, strips terrain, takes connected components, removes duplicates, and crops to a near-tight bounding box; it also rejects candidates whose main body lies below the surface, so underground dwellings fall outside the house scope this dataset covers (a limitation of the dataset; full rules in Appendix B). The pipeline yielded 10,228 admissible candidates. We release the 4,030 builds labeled house or other_structure. The rest (labeled unclear, non-structure or unfinished, plus 143 builds withheld after a content screen) are not released. The median size is $15 \times 15 \times 15$ cells, the 90th percentile $38 \times 36 \times 36$. Labels are binary, assigned per build by a single annotator; the criterion is holistic, covering appearance and residential purpose; the label is therefore not a function of the measures, which read space alone.

A known limitation is that the data are heavily concentrated: the largest cluster holds 41.7% of the houses and 38.6% of the negatives (top three: 79.6% and 71.8%), so every confidence interval on the human and control rows resamples **map files**, not individual samples; CIs for the generators resample samples per source (Appendix D).

Provenance and takedown. This dataset rests on the work of anonymous community builders; per-world attribution ships with the released set (ATTRIBUTION.md). The dataset is released under CC BY-NC 4.0; uploader-declared source licenses are recorded per world in ATTRIBUTION.md (self-reported, not legal determinations). Individual builders cannot be traced one by one (a structural limit of community map corpora). The geometry-only export (a block vocabulary and a state grid) contains no sign text, book contents or player names, so the released data carries no personally identifiable information; takedown requests are honored via the repository's issue form or by e-mail to the maintainer, and removal ships with the next release. Code (`structgauge`, Apache-2.0): https://github.com/KyleLin0927/structgauge; dataset: https://huggingface.co/datasets/jenkai-lin/VoxHouse3D.

Table 1: Discrimination on VoxHouse3D (house vs non-house).

Measure	AUC (house vs non-house)	95% CI (cluster bootstraps)
Shelter	**0.8681**	[0.8421, 0.8805]
Access	**0.8424**	[0.8255, 0.8653]
Size-only baseline (solid fraction)	0.6550	[0.6081, 0.6895]
Size-only baseline (log block count)	0.6275	[0.5808, 0.6683]
Isovist openness [1]	0.7643	[0.7199, 0.7947]

Table 2: Permutation control: 400 buildings, blocks re-placed at random with types and counts unchanged. Dropped: share of builds whose score falls after shuffling; Δ: median change (shuffled $-$ original). The range boundaries are scanned in Appendix C.2.

Score range	Shelter dropped	Shelter Δ	Access dropped	Access Δ
Low $[0, 0.3)$	25%	$+0.055$	27%	$+0.040$
Mid $[0.3, 0.7)$	89%	-0.235	92%	-0.253
High $[0.7, 1]$	98%	-0.460	100%	-0.500

5 Validity

The two measures make their house-quality claim in the low direction only, so the weight of the validity argument falls on that same side [13].

The discrimination is not produced by size or provenance (Table 1). Confidence intervals use a cluster bootstrap whose resampling unit is the **map file**, since buildings in one map file share players and period: each resample draws whole map files rather than individual buildings [15, 22]. Nor is the discrimination an artifact of cluster composition: computing the AUC [14] per map file and then reweighting gives **0.8816** for shelter and **0.8367** for access, close to the pooled figure; dropping the largest map file (40% of the data) moves the two by only -0.0012 and $+0.0106$. Stratifying within size deciles and reweighting gives 0.8528 and 0.8273, so size confounding is ruled out as well.

5.1 Low-score census and identifiable-condition rate

We audited the low tail of the house class in the release set as a **census**, not a sample: all 119 buildings with either measure < 0.2 (60 in the low-shelter group, 59 in the low-access group), under a rubric fixed before judging. The flagged condition is present in the geometry of **59 of 60** low-shelter buildings and **59 of 59** low-access buildings; the measure itself produced one false alarm on shelter (`VH3D-w03-3b16f218`, 0.187) and none on access. One qualitative observation: most failures in the low-access group are invisible in renders, which is why adjudication required a 3D inspector; visual inspection cannot substitute for the measure.

The rubric also records what kind of building carried the condition. Real defects (15 low-shelter, 44 low-access): a hole or an unfinished wall; no entrance, a blocked doorway, or floors without stairs. Intended openness (39 and 9): pavilions, terraces and other builds open by design or too open, plus builds entered only through a redstone mechanism that the static geometry records as closed; the opening is real; the label records intent. Data-preparation defects (5 and 6): segmentation gaps at entrances and thin-block semantics, the rules deciding what counts as solid or passable (Section 6.1).

The adjudicator is an author of this paper, so the audit is not independent and remains open to third-party re-adjudication; the 0.2 was likewise fixed in advance.

5.2 Permutation control and criterion validity

As a falsification experiment, we randomly re-place the blocks of each building and measure again [21]. If the measures only read "which blocks are present", the scores should not move. Almost every high-scoring building drops (Table 2); the builds whose scores do not drop are concentrated at the low end, where the score is already saturated, which is not evidence that the measure is blind to arrangement.

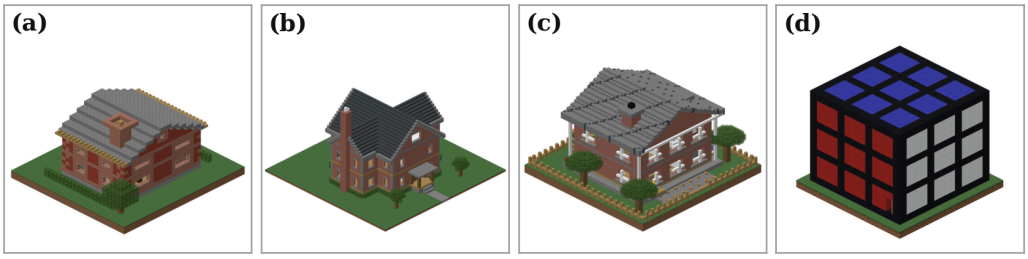


Figure 4: An intact exterior does not mean a correct structure. (a)–(c) LLM-generated builds: both judges give 9.5–10/10 on every question (house, shelter, access), yet measured access is 0.099 / 0.070 / 0.218; the defect is inside. (d) A Rubik's cube with a door (VH3D-w01-ca78cf9d): enclosed and enterable, so the measures read 0.953 / 0.908 and the judges agree (shelter 10, access 7.5), yet annotator and judges call it non-house (0/10). The measures say only that the structure works; house identity sits above the structural level, where a visual judge is competent.

Table 3: Alarm rates: share of each source below the human-house 10th-percentile threshold (shelter 0.3748, access 0.4526), with 95% per-build bootstrap CIs; the human rate is 10% by construction, and the two reference rows carry no CI. Either = below threshold on at least one measure.

Source	n	Shelter [95% CI]	Access [95% CI]	Either
Human house (reference)	2,007	10.0%	10.0%	18.1%
Same-provenance non-house	2,023	70.4%	65.5%	87.9%
Creative Agents (image-to-voxel reconstruction) [32]	1,000	**90.5%** [88.7, 92.3]	66.9% [63.9, 69.6]	94.4%
Build-with-Bombs (voxel diffusion) [41]	2,000	**88.6%** [87.2, 90.0]	20.5% [18.8, 22.3]	92.2%
MSG (autoregressive completion) [42]	300	**86.0%** [82.0, 90.0]	47.3% [41.7, 53.3]	92.3%
Direct LLM (pooled; Appendix D.1) [40, 43, 44]	178	27.0% [20.8, 33.7]	**53.4%** [45.5, 60.7]	73.6%

5.3 Comparison with visual judges

Why not simply use a visual judge? We ask two commercial multimodal judges about 100 houses and 100 non-house structures from the released set (setup, verbatim questions and disclosures in Appendix C.4). At recognizing houses they are excellent (AUC 0.9255 [0.8836, 0.9608]); the measures do not compete at that level: house identity is decided above the structural level. On the two conditions the measures do target, the judges provide no substitute signal: within the house class their access scores track the measured values only weakly ($\rho = +0.12$, 95% CI $[-0.09, 0.38]$), and their shelter scores likewise ($\rho = +0.27$ $[0.05, 0.45]$; a third of the scores sit at the top mark). The reason is structural: a judge sees the shell and must guess the interior that the measures compute (Figure 4; interior-view judges remain untested; generator-build results in Appendix C.4).

5.4 The measures applied to current generators

These experiments are also the first run of the protocol: the alarm rates come from the released `structgauge bench`, so a new generator run against the same pinned threshold is directly comparable with ours. We apply the same measures to four building-task generation sources (details and scoring rules in Appendix D) and report only the one-sided quantity, the fraction of each source below the human-house 10th-percentile threshold (Table 3; the human rate is 10% by construction).

Every generator cell is at least twice the human rate, and this is insensitive to the threshold: with 5th/25th-percentile thresholds, 7/8 or 8/8 cells remain at or above twice the human rate, and across absolute thresholds 0.15–0.45 the ratio never approaches 1 (scan in Appendix D.2). More importantly, the **failure modes diverge**: the image-reconstruction, voxel-diffusion and autoregressive-completion sources sit at 86–91% on shelter but 21–67% on access: they build shells that cannot keep out wind and rain (Build-with-Bombs, at 21%, is walkable but not weatherproof); direct LLM generation is the reverse, 53% on access against 27% on shelter: it builds boxes one cannot enter. Same measures, same threshold; different families fail on different embodied properties.

These columns are alarm rates, not quality scores; the guarantee is one-sided (as with SWE-bench): a source's alarm rate can improve, but 0% does not mean its buildings are good houses.

6 Discussion and limitations

6.1 What the measures claim

A high score asserts only two structural properties: the interior Ω mass is enclosed along the five axis directions (shelter), or a walkable path exists from relatively outdoor space to relatively indoor space (access). It is not a quality claim: the measures test necessary conditions of usability, and passing them says nothing about aesthetics or style, nor about house identity (Figure 4d). We therefore do not rank individual buildings (AUC is interpreted only as group-level discriminant validity), do not use the high end as a reward signal, and do not fuse the two into one score; the Section 5.4 threshold is a batch-wide statistical cut, not a per-build pass mark.

The low direction claims that the building fails a necessary condition of a usable house, checkable per building; for a structure that is open by design, the missing cover is intended: the label records the intent, the measure records only the function. Low shelter identifies no roof or open sides; low access identifies a build that cannot be entered (sealed, fragmented walkable components, or an interior field that is low throughout). Limitations remain: water elevators, non-integer-height passages and piston-operated entrances are unmodeled, so access underestimates what a player can reach; trapdoors read as walls and carpets as roofs, so shelter undercounts.

The labels come from a single annotator; three independent annotators (non-authors, blind to labels and measures) reach 93/98 majority-vote agreement ($\kappa = 0.89$; Appendix B), with disagreements concentrated on how leniently the house boundary is drawn. But not all of the validity argument rests on labels: controlled geometry and the judge–measure correlation use none, while the criterion-validity AUC, the census selection, and the P10 threshold do.

6.2 Dependence on γ

Over $\gamma \in \{0.80, 0.85, 0.90, 0.95\}$ the measures move strictly monotonically in opposite directions (AUC: shelter $0.8709 \to 0.8631$, access $0.7990 \to 0.8644$), so no γ favors both; we do not select γ by AUC [3]; we report $\gamma = 0.90$. Shelter's ordering barely moves while access's moves more (Spearman against $\gamma = 0.90$ at least 0.998 vs 0.979), a known property of discounted scores [33].

The absolute values move the other way round: γ shifts shelter's values by 34.2% (28% on 3D-Craft [31], a second corpus) and access's by 7.0% (8.8%). The absolute-threshold alarms therefore move with γ, so a release must freeze and publish both the thresholds and γ.

6.3 Gaming a single axis

Any one-sided measure read alone invites gaming toward its high end. The sealed box is the simplest demonstration (shelter 1.0000, access 0.0000): with shelter as the sole objective the cheapest solution is a doorless shell (Appendix E). A text-to-building LLM pipeline exhibits this unprompted ($n = 30$, Appendix D.1): median shelter 0.9590 with AUC fully inverted to 0.0855, yet access 0.0265 with AUC 0.9907; one measure is saturated while the other catches the defect; they must be judged together.

7 Conclusion

We turn the two questions into computable, falsifiable one-sided measures, with a dataset of same-provenance negatives and a validity audit: a low score means the building fails a necessary condition of a usable house; a high score asserts only the structural property. In use, the measures are rejection gates rather than rewards to maximize (Section 6.3): in agent training a low score is a penalizable, identifiable defect; in benchmarks, a source is characterized by its alarm rates against the frozen human reference threshold, reported on both axes (Section 5.4). This paper delivers the instrument and the evidence for its validity.

References

[1] C. Salge, J.-B. Hervé. Automated Isovist Computation for Minecraft. FDG 2022.

[2] J.-B. Hervé, C. Salge. Comparing PCG Metrics with Human Evaluation in Minecraft Settlement Generation. FDG 2021.

[3] S. Steegen, F. Tuerlinckx, A. Gelman, W. Vanpaemel. Increasing Transparency Through a Multiverse Analysis. Perspectives on Psychological Science, 2016.

[4] M. L. Benedikt. To Take Hold of Space: Isovists and Isovist Fields. Environment and Planning B, 1979.

[5] A. Turner, M. Doxa, D. O'Sullivan, A. Penn. From Isovists to Visibility Graphs. Environment and Planning B: Planning and Design, 2001.

[6] K. D. Lathrop, B. G. Carlson. Discrete Ordinates Angular Quadrature of the Neutron Transport Equation. Los Alamos Scientific Laboratory Report LA-3186, 1964.

[7] S. Zhukov, A. Iones, G. Kronin. An Ambient Light Illumination Model. Rendering Techniques '98, 1998.

[8] L. Zheng et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS 2023 Datasets and Benchmarks Track.

[9] P. Wang et al. Large Language Models are not Fair Evaluators. ACL 2024.

[10] M. Heusel et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. NIPS 2017.

[11] G. Smith, J. Whitehead. Analyzing the Expressive Range of a Level Generator. PCGames 2010.

[12] O. Withington et al. On the Evaluation of Procedural Level Generation Systems. FDG 2024.

[13] L. J. Cronbach, P. E. Meehl. Construct Validity in Psychological Tests. Psychological Bulletin, 1955.

[14] J. A. Hanley, B. J. McNeil. The Meaning and Use of the Area under a Receiver Operating Characteristic (ROC) Curve. Radiology, 1982.

[15] N. A. Obuchowski. Nonparametric Analysis of Clustered ROC Curve Data. Biometrics, 1997.

[16] J.-B. Hervé et al. An Examination of the Hidden Judging Criteria in the Generative Design in Minecraft Competition. IEEE Transactions on Games 16 (3):635–644, 2024.

[17] Z. Wei et al. MineAnyBuild: Benchmarking Spatial Planning for Open-world AI Agents. NeurIPS 2025 Datasets and Benchmarks Track.

[18] C. Salge et al. Generative Design in Minecraft (GDMC): Settlement Generation Competition. FDG 2018.

[19] J. Gray et al. CraftAssist: A Framework for Dialogue-enabled Interactive Agents. arXiv:1907.08584, 2019.

[20] T. R. Oke. Canyon Geometry and the Nocturnal Urban Heat Island. Journal of Climatology, 1981.

[21] M. T. Ribeiro et al. Beyond Accuracy: Behavioral Testing of NLP Models with CheckList. ACL 2020.

[22] A. C. Cameron, J. B. Gelbach, D. L. Miller. Bootstrap-Based Improvements for Inference with Clustered Errors. The Review of Economics and Statistics, 2008.

[23] L. Fan et al. MineDojo: Building Open-Ended Embodied Agents with Internet-Scale Knowledge. NeurIPS 2022 Datasets and Benchmarks Track.

[24] P. Cousot, R. Cousot. Abstract Interpretation: A Unified Lattice Model for Static Analysis of Programs by Construction or Approximation of Fixpoints. POPL 1977.

[25] B. Livshits et al. In Defense of Soundiness: A Manifesto. Communications of the ACM, 2015.

[26] E. Perez et al. Red Teaming Language Models with Language Models. EMNLP 2022.

[27] C. M. Goral et al. Modeling the Interaction of Light Between Diffuse Surfaces. SIGGRAPH 1984.

[28] H. Tam et al. SceneEval: Evaluating Semantic Coherence in Text-Conditioned 3D Indoor Scene Synthesis. WACV 2026.

[29] Y. Yang et al. PhyScene: Physically Interactable 3D Scene Synthesis for Embodied AI. CVPR 2024.

[30] J. Yang et al. Thinking in Space: How Multimodal Large Language Models See, Remember, and Recall Spaces. CVPR 2025. arXiv:2412.14171.

[31] Z. Chen et al. Order-Aware Generative Modeling Using the 3D-Craft Dataset. ICCV 2019.

[32] P. Cai, C. Zhang, Y. Fu, H. Yuan, Z. Lu. Creative Agents: Empowering Agents with Imagination for Creative Tasks. UAI 2025, PMLR 286:471–496. arXiv:2312.02519.

[33] M. Bressan, E. Peserico. Choose the Damping, Choose the Ranking? Journal of Discrete Algorithms, 2010.

[34] Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli. Image quality assessment: from error visibility to structural similarity. IEEE Trans. Image Process. 13(4):600–612, 2004.

[35] P. Jayannavar et al. BAP v2: An Enhanced Task Framework for Instruction Following in Minecraft Dialogues. Computational Linguistics, 2026. arXiv:2501.10836.

[36] C. E. Jimenez et al. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? ICLR 2024. arXiv:2310.06770.

[37] M. Deitke et al. ProcTHOR: Large-Scale Embodied AI Using Procedural Generation. NeurIPS 2022. arXiv:2206.06994.

[38] B. Chen et al. SpatialVLM: Endowing Vision-Language Models with Spatial Reasoning Capabilities. CVPR 2024. arXiv:2401.12168.

[39] M. Savva et al. Habitat: A Platform for Embodied AI Research. ICCV 2019. arXiv:1904.01201.

[40] S. Hu, Z. Huang, C. Hu, J. Liu. 3D Building Generation in Minecraft via Large Language Models. IEEE CoG 2024. arXiv:2406.08751.

[41] T. Barnes. Build with Bombs: a Minecraft mod for in-game house generation by voxel diffusion. https://buildwithbombs.com; https://github.com/timothy-barnes-2357/Build-with-Bombs. Accessed July 2026.

[42] mmmfrieddough. Minecraft Schematic Generator: transformer-based structure completion for Minecraft. https://github.com/mmmfrieddough/minecraft-schematic-generator. Accessed July 2026.

[43] CyniaAI. BuilderGPT: an LLM-based structure generator for Minecraft. https://github.com/CyniaAI/BuilderGPT. Accessed July 2026.

[44] A. Alam. MineBench: benchmarking spatial reasoning through Minecraft-style voxel construction. https://minebench.ai; https://github.com/Ammaar-Alam/minebench. Accessed July 2026.

Table 4: Rejection categories of the segmentation pipeline.

Category	Count	Meaning
ok (enters the annotation pool)	10,228	Passes all gates
reject_too_small	3,846	Artificial block count below the minimum
reject_worldgen_dungeon	2,236	Dungeon signature (spawner bricks / mossy floor)
reject_underground	2,308	Main body below the surface
reject_monotone	1,646	Insufficient block diversity (single-material mass)
reject_sparse	1,523	Artificial density below threshold
reject_worldgen_structure	554	Overlaps a worldgen structure box
reject_buried	593	Buried by terrain

A Method details: ray traversal, symmetry, and cost

(Supports main text Section 3.1; notation follows the main text.)

Ray traversal and rotation invariance. From every cell, a ray–voxel walk (Amanatides–Woo DDA) proceeds along the 24 directions of a level-symmetric (S_N-type) quadrature [6] to the first hit; the direction set is fixed and small because Ω is the exact solution of a sparse linear system built from it (main text Section 3.1), and symmetric because octahedral invariance requires it. Every such direction has two components of equal absolute value, so the ray crosses a cell edge or corner at every step, and the tie rule is part of the measure's definition. We use the geometric supercover rule (on a tie, advance both axes simultaneously and check the grazed intermediate cells): it does not depend on axis order, so the field is invariant under all 48 octahedral symmetries by construction (an axis-order rule, which always advances the lowest-numbered axis on a tie, singles out one axis and loses the symmetry), and it does not pass through zero-width diagonal seams where two solid blocks meet only along an edge. We verify the symmetry per sample: 24 released buildings, sampled at equal size quantiles, each checked under all 48 group elements; all pass, with worst NMAE (normalized mean absolute error) 5.0×10^{-16} and worst $\max|\Delta|$ 3.2×10^{-15}; the criterion also requires that cells carrying a value map onto the original set after rotation.

Cost. The full pipeline (geometry preparation, solving the field, and both measures) over all 4,030 released buildings takes 37.3 minutes total on a single core: per-building median 0.083 s, maximum 182.2 s (a heavy-tailed distribution; large buildings dominate the total).

B Dataset construction: segmentation rules and rejection categories

(Supports main text Section 4.)

The segmentation pipeline runs in order: exclude vanilla worldgen structures using the world's own structure records; strip terrain and vegetation; take 6-connected components of artificial blocks, stitched across region boundaries and keeping up to two layers below ground level; remove stray fragments, but keep a component intact when its secondary pieces make up at least 20% of it; deduplicate, counting mirrored copies as identical and merging two builds when the larger of their two directed overlaps reaches 0.85, or 0.78 when their block palettes also agree (Jaccard similarity 0.5); crop to a near-tight bounding box.

The released set was frozen at 4,030 buildings after manual annotation of the ok pool (house / other_structure). Removing the 110 buildings involved in cross-label bounding-box overlaps moves shelter AUC from 0.8681 to 0.8692 and access AUC from 0.8424 to 0.8456.

Inter-annotator agreement. Three independent annotators (non-authors, blind to labels and measures) each labeled the same stratified sample of 100 builds (30 clear-house / 30 clear-other / 40 deliberately oversampled boundary). Cohen κ against the released labels: 0.9555 / 0.7750 / 0.6841; pairwise κ among the three: 0.7255 / 0.8220 / 0.6604. **Majority vote agrees with the released labels on 93/98 ($\kappa = 0.8943$)**; 2 items lack a majority; all 5 flips relabel other_structure as house, the lenient direction, and sit in the boundary stratum: the disagreement is leniency about "what counts as a house", not noise, and all three annotators find the released house labels clean (28–29/30 in clear-house). The boundary stratum is oversampled, so these figures understate population agreement.

Table 5: Per-building readings and manually reviewed defect descriptions.

Ex	Side	Defect type	public_id	Shelter	Access	Defect description (manual review)
1	access	doorway impassable	VH3D-w03-db8ac166	0.8587	0.0199	entrance sealed with iron blocks
2	access	floors disconnected	VH3D-w10-91f051e0	0.6799	0.0966	upper floor unreachable, no stairs
3	access	no entrance	VH3D-w07-62177778	0.7402	0.0376	no entrance at all
4	access	no entrance	VH3D-w07-c48d3305	0.7371	0.0630	no entrance at all
5	access	doorway impassable	VH3D-w03-c50e648b	0.8157	0.1444	hidden gap at the entrance
6	access	doorway impassable	VH3D-w03-8c58696d	0.8159	0.0658	doorway sits high up, unreachable from the ground
7	shelter	enclosure breach	VH3D-w16-78100c5b	0.1417	0.2106	roof hole pierces three floors
8	shelter	no interior	VH3D-w25-41d19da1	0.0000	0.3090	normal exterior, no interior at all
9	shelter	open by design	VH3D-w23-ecfc4a80	0.1644	0.3831	looks like a house; is an unenclosed gazebo
10	shelter	no roof	VH3D-w16-bb71027e	0.0025	0.6375	no roof
11	shelter	enclosure breach	VH3D-w08-666f655f	0.1318	0.5409	front view fully normal; large rear breach
12	shelter	enclosure breach	VH3D-w03-12174a77	0.0977	0.4856	wall hole, visible only up close; looks intact from a single view

C Validity details

C.1 Named low-score example set

(Supports main text Section 5.1.) The example set was confirmed by the annotator through per-building manual review of low-scoring samples (Figure 5). The example set is drawn from the low-score census of main text Section 5.1 (any measure < 0.2, 119 buildings): 12 examples, each manually reviewed build by build, each also re-adjudicated in the census. The rear breach (example 11) and the no-entrance cases (examples 3 and 4) provide direct evidence of defects a single-view photograph cannot show. The 12 examples come from 7 worlds; this set is a defect catalog, not a representative cross-world sample. The carpet-roof cases (thin blocks as roofs, so the cover is not counted and shelter collapses) comprises 4 buildings in the census of main text Section 5.1: VH3D-w10-0687bc5b, VH3D-w08-affe39d5, VH3D-w08-758a1290, VH3D-w10-d02f909b; all are recorded there as data-preparation defects.

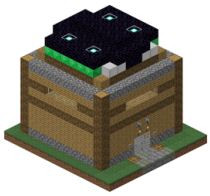
C.2 Permutation control: sensitivity to the range boundaries

(Supports main text Section 5.2 and Table 2.)

Sensitivity to the range boundaries. Scanning the same 400-building paired-permutation data: for the upper bound of the low range, $\tau_{lo} \in \{0.15, \ldots, 0.45\}$, the fraction that is lower after permutation runs 0.151→0.335 (shelter) and 0.135→0.431 (access), staying $\leq$0.5 throughout; for the lower bound of the high range $\tau_{hi} \in \{0.50, \ldots, 0.80\}$, shelter runs 0.966→1.000 and access 0.996→1.000, staying $\geq$0.9 throughout, so the conclusion is insensitive to the thresholds.

C.3 Measure properties

Five properties of the measures are validated on the released data; the measure values they rest on can be recomputed by a third party from the released data and code. The two measures are not redundant: within the house class their Spearman ρ is +0.3267. They read space, not palette: a bag-of-blocks baseline reaches AUC 0.7306, against 0.8681 for shelter and 0.8424 for access. Discrimination is not driven by size: the paired Δ against the strongest size-only baseline (fill_ratio) is positive throughout, +0.2140 for shelter and +0.1884 for access. Conclusions are robust to the discount parameter γ: rankings change little under the scan (Spearman $\geq$0.977 against $\gamma = 0.90$), while shelter values shift by 34.2% and access by 7.0% on the released set (28% and 8.8% on the second corpus). Finally, the measures read arrangement, not just composition: shuffling the positions of the same blocks (same types, same counts) at random, the score drops for 98.3% (shelter) and 100% (access) of high-scoring ($\geq$0.70) buildings.



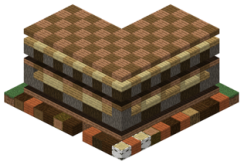
1 VH3D-w03-db8ac166
shl=0.859 acc=0.020
iron blocks seal the doorway



2 VH3D-w10-91f051e0
shl=0.680 acc=0.097
no stairs to 2nd floor



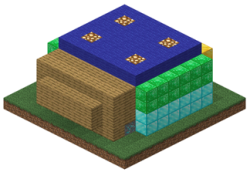
3 VH3D-w07-62177778
shl=0.740 acc=0.038
no doorway at all



4 VH3D-w07-c48d3305
shl=0.737 acc=0.063
no doorway at all



5 VH3D-w03-c50e648b
shl=0.816 acc=0.144
hidden gap at the entrance



6 VH3D-w03-8c58696d
shl=0.816 acc=0.066
door high up, unreachable



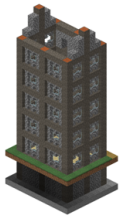
7 VH3D-w16-78100c5b
shl=0.142 acc=0.211
roof hole pierces floors 1-3



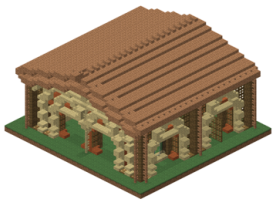
8 VH3D-w25-41d19da1
shl=0.000 acc=0.309
no interior at all



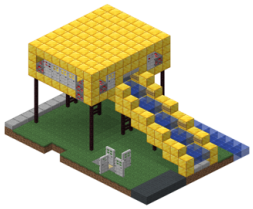
9 VH3D-w23-ecfc4a80
shl=0.164 acc=0.383
looks like a house; is a gazebo



10 VH3D-w16-bb71027e
shl=0.003 acc=0.637
no roof



11 VH3D-w08-666f655f
shl=0.132 acc=0.541
rear breach, hidden from the front



12 VH3D-w03-12174a77
shl=0.098 acc=0.486
wall hole, visible only up close

Figure 5: Twelve true positives that "look fine" from outside: examples 1–6 on the access measure and 7–12 on the shelter measure (a roof hole through all floors, no interior, gazebo, no roof, rear breach, hidden breach); values in Table 5.

C.4 Visual-judge comparison details

(Supports main text Section 5.3.)

Setup. Two judges (gpt-4o-2024-08-06 / gemini-2.5-flash, thinkingBudget=0), temperature 0, max tokens 300, JSON schema enforced (reason first, score after), four independent view renders per sample (width 640, proportional height). Each of the three questions below is asked in its own call, so no score can influence another; the fixed seed applies to the GPT judge only (the Gemini API exposes no seed). Samples: 100 houses from the released set and 100 other_structure negatives (4×4 shelter$\times$access quartile strata, $\leq$10 per world; the 178 generator builds of Appendix D.1, and the two houses of Figure 1.

Verbatim question text (each question is a separate call; every call starts with the common preamble):

> You are evaluating a voxel structure from the game Minecraft. The four images below are FOUR VIEWS OF THE SAME SINGLE STRUCTURE, rotated 90 degrees apart. Judge ONLY what is asked. Do NOT judge beauty, style, or creativity.
>
> [shelter] Question (shelter_score, 0-10): If a player stood inside this structure during a storm, how well would it physically block rain from above and wind/projectiles from the four sides? 0 = completely exposed (no roof and no walls); 5 = partial cover (e.g., roof but open sides, or walls but no roof); 10 = fully enclosed cover on top and all four sides. Answer in JSON only.
>
> [access] Question (access_score, 0-10): Could a player walk into and move around inside this structure? Consider: is there a usable entrance (door or opening reachable from the ground), and is there walkable interior space behind it? 0 = impossible to enter (sealed, or no interior at all); 5 = enterable but cramped or awkward (e.g., must jump/climb, single tiny room); 10 = easy entrance and freely walkable interior. Answer in JSON only.
>
> [house] Question (is_house, 0-10): Is this a house, that is, a structure built as a dwelling with an interior meant to be lived in? Judge appearance and residential purpose together. 0 = clearly not a house (terrain, random blocks, decoration, vehicle, statue); 5 = ambiguous; 10 = clearly a house. Answer in JSON only.

Results. The is-house AUC is 0.9255 [0.8836, 0.9608] (map-file cluster bootstrap B=2000; gpt 0.9096 / gemini 0.8926); the condition-tracking correlations are in Table 6.

Table 6: Judge scores versus measured values (Spearman ρ, two-judge average). All rows come from a single run with the setup above. Pooled values are an artifact of between-class separation, the pooling lesson of Appendix D.1. The last two columns give the share of judge scores sitting at the top mark of 10, i.e., the ceiling.

Samples	n	ρ shelter	ρ access	shelter at 10	access at 10
House class	100	$+0.274$	$+0.117$	33%	33%
Non-house class	100	$+0.486$	$+0.334$	12%	10%
Generator builds	178	-0.101	$+0.352$	55%	37%

D Characterizing generators

(Supports main text Section 5.4 and Section 6.3.)

The comparison semantics of alarm rates are one-sided: within one axis the severity of generator failures can be compared (as SWE-bench compares failure rates), but the two axes are never merged into a single score, and a low alarm rate does not imply "better overall" (the high end makes only structural claims; main text Section 6.1). The generators' task settings differ somewhat (in-game on-site generation, partial-structure completion, instruction-following construction, image reconstruction, etc.); the measures read the produced geometry itself. The Creative Agents row consists of reconstructions by its image-to-voxel component from renders of real buildings (not generation from scratch); the geometry carries the structure of human builds, so its rows measure reconstruction rather than free generation. Samples with no interior space at all are scored 0 by convention by both measures under the pinned semantics and count in the denominators; exactly 1 of Build-with-Bombs' (BwB) 2,000 samples in these tables is such a case. Sample sizes and cluster

Table 7: LLM sub-source split. The six cells share one set of 30 prompts; the two $n = 29$ cells each exclude one invalid output (response with no parseable block matrix) or one duplicate output. Reference = VoxHouse3D house ($n = 2{,}007$, the same reference as Table 3; values computed with the same released pipeline and version. The text-to-building pipeline is a faithful reproduction of T2BM, the LLM building pipeline of [40]; builder-prompt v2 = BuilderGPT [43]; task-suite prompt = MineBench [44]; the three pipelines share the same two models: GPT = gpt-5.6-sol, Gemini = gemini-3.1-pro-preview, temperature 1.0.

Sub-source (pipeline × model)	n	Shelter med.	Shelter AUC	Access med.	Access AUC
text-to-building × Gemini	30	0.9590	(inv.) **0.0855**	0.0265	**0.9907**
text-to-building × GPT	29	0.6219	0.5985	0.1751	0.9404
builder-prompt v2 × Gemini	29	0.5945	0.6886	0.7587	0.6181
builder-prompt v2 × GPT	30	0.4882	0.7756	0.6561	0.7387
task-suite prompt × GPT	30	0.4042	0.8473	0.4233	0.8781
task-suite prompt × Gemini	30	0.3144	**0.8640**	0.4901	0.8365

structures differ across sources; all alarm counts are computed with the pinned constants $\gamma = 0.90$ (γ dependence: main text Section 6.2). Source sizes are unequal (generators 178 to 2,000; human and control rows 2,007/2,023): an equal-n subsampling check ($n = 178$ per source × 200 draws, fixed seed) shifts the full-batch rates by at most 0.31 percentage points on average, so unequal n does not drive this section's conclusions; a single equal-n draw would instead introduce roughly ± 3 to 8 percentage points of sampling noise, so the tables keep full-batch values; the CIs convey the precision.

D.1 Splitting the LLM sub-sources: the pooled figure hides three regimes

(Supports main text Section 6.3.)

Shelter AUC spans 0.0855 to 0.8640 and access 0.6181 to 0.9907, and the two extremes come from the same language model under different prompt pipelines, so at least three distinguishable failure modes exist in this LLM output: too sealed (text-to-building × Gemini, the solid-shell score-inflation instance of Appendix E), too open (the task-suite pipeline), and mixed in between. The 95% CIs of the two extremes: the shelter inversion is 0.0855 [0.0226, 0.1596] and the same batch's access is 0.9907 [0.9812, 0.9964]. Per pipeline: the prompt-pipeline effect exceeds the model effect; the same Gemini occupies both shelter-AUC extremes under text-to-building and task-suite (0.0855 and 0.8640); under the same text-to-building pipeline the two models show opposite pathologies, Gemini's builds are nearly sealed (shelter median 0.9590) while GPT's shelter scores are the hardest to distinguish from human builds (AUC 0.5985); the builder-prompt v2 pipeline sits roughly in the middle; most of the task-suite pipeline's builds are open (shelter medians 0.31 to 0.40; both AUCs around 0.84).

D.2 Alarm-rate sensitivity to an absolute threshold

(Supports main text Section 5.4.) Table 8 replaces the P10 threshold with absolute cuts from 0.15 to 0.45. At every cut the worst generator stays 5.9–33.2× the human rate on shelter and 6.9–11.5× on access, never approaching 1: the gap between human and generated buildings depends neither on the P10 cut nor on any single absolute threshold.

E Controlled shell sequence and solid-shell score inflation

(Supports main text Section 6.3.) Fully synthetic, no external corpus; a single $11 \times 7 \times 11$ shell with enclosure elements added or removed one at a time ($\gamma = 0.9$; per-state block collision semantics taken from the game's official data files; the states are rendered in Figure 6).

Shelter's largest movements (Table 9) come from adding the roof or the walls; access's largest movement comes from opening the door: the argmax of the two measures falls on different edits.

Table 8: Alarm rates of human houses and of the worst generator at absolute thresholds 0.15–0.45; the P10 thresholds (0.3748 / 0.4526) fall inside this range. Ratio = worst generator divided by human.

Threshold	Human shelter	Worst gen. shelter	Ratio	Human access	Worst gen. access	Ratio
0.15	2.0%	66.0%	33.2×	2.4%	27.5%	11.5×
0.20	3.0%	74.4%	24.9×	2.9%	31.5%	10.7×
0.25	4.3%	80.3%	18.7×	3.8%	34.8%	9.1×
0.30	6.7%	85.1%	12.6×	4.8%	40.3%	8.3×
0.35	8.6%	88.6%	10.3×	6.1%	49.0%	8.1×
0.40	11.6%	91.8%	7.9×	7.6%	58.8%	7.8×
0.45	15.8%	93.8%	5.9×	9.6%	66.8%	6.9×

Table 9: The controlled shell sequence. The first row has no interior mass; its zero scores are assigned by convention.

Geometry	Shelter	Access	Defect flagged	Solid cells
Fully solid (no interior)	0.0000	0.0000	none	847
Walls, no roof (walls fully closed)	0.0000	0.1520	no roof	361
Roof, no walls (pavilion)	0.0000	0.0981	open sides	242
Walls and roof, no door (sealed box)	**1.0000**	**0.0000**	sealed interior	442
Walls and roof plus one door	0.9972	0.4273	none	442
Walls, no roof, plus one door (with-door control)	0.0000	0.1862	no roof	361

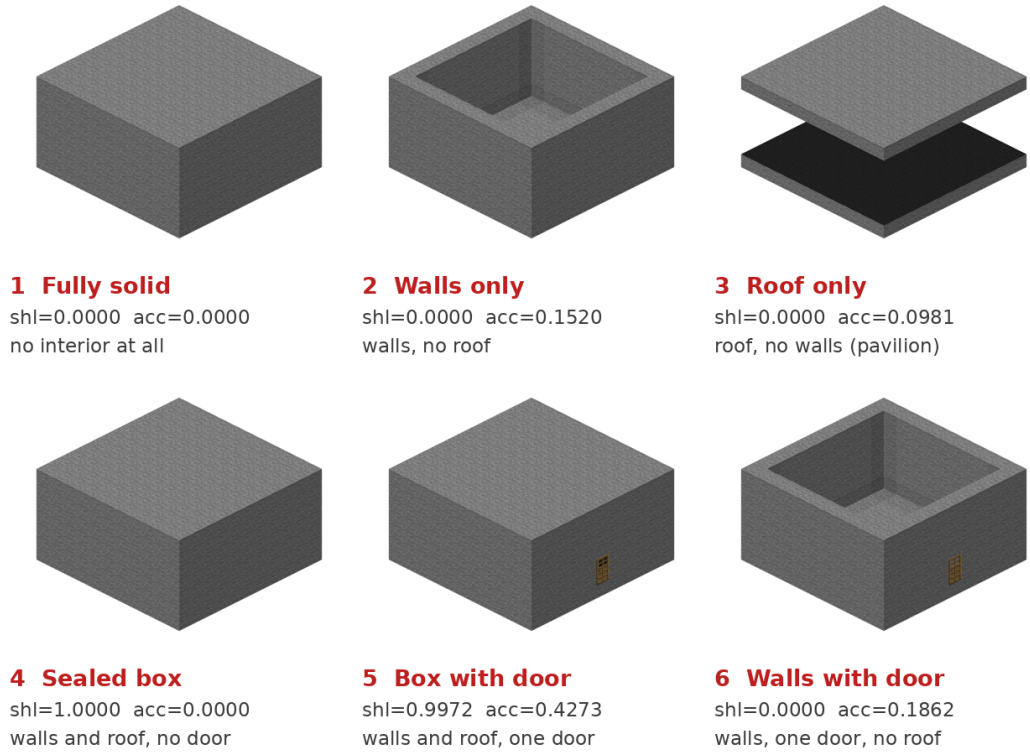


Figure 6: The six states of the controlled shell sequence of Table 9, rendered from the same $11 \times 7 \times 11$ geometry, with their shelter and access readings.